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Lithium Metal Batteries

Asymmetrically Fluorinated Alkoxysilane Single-Solvent Electrolytes Enable High-Voltage and Long-Cycling Lithium Metal Batteries

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Abstract: High-voltage lithium metal batteries with nickel-rich oxide cathodes ($LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$, NCM811) represent one of the most promising approaches to achieve high energy density up to 500 Wh kg^{-1}. However, severe interfacial side reactions occur at both NCM811 cathode and lithium anode at ultrahigh voltages (>4.6 V). To address these issues, various electrolytes have been developed, but they still suffer from electrolyte decomposition, leading to moderate voltages and insufficient cycling. Herein, we introduce (3,3,3-trifluoropropyl)trimethoxy silane (TTMS) as an asymmetrically fluorinated single solvent, which incorporates both strongly solvating (−OCH_3) and weakly solvating (−CF_3) groups. The designed 2.1 mol L^{-1} (M) LiFSI/TTMS electrolyte achieves excellent compatibility with both NCM811 cathode and Li metal anode due to its unique anion-dominating solvation structures and inorganic-rich interphase formation. Consequently, it enables stable cycling in the Li||NCM811 battery at an ultrahigh voltage of 4.8 V, with 84.5% capacity retention after 300 cycles. Even under more aggressive conditions, including high temperature (60 °C) and anode-less configuration (N/P ratio = 1.76), the Li||NCM811 battery exhibits remarkable capacity retention (>80%) over 300 cycles. This work underscores the effectiveness of electrolyte engineering for developing ultrahigh-voltage and long-cycling battery systems.

Introduction

There is an increasing demand to develop high-energy batteries for emerging applications, such as long-range electric vehicles, portable electronics, and drones.[1–3] Currently, lithium-ion batteries (LIBs) struggle to provide moderate energy density of 150–250 Wh kg^{-1}, due to the limited capacity in the metal oxide cathode and graphite anode. To increase energy density, it is imperative to develop high-capacity electrode materials with large potential gaps. On the cathode side, $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM811) is a highly competitive material due to its high theoretical capacity (~280 mAh g^{-1}) and high operation voltage (>3.7 V). However, its practical capacity is limited to 190−200 mAh g^{-1} at a cutoff voltage of 4.3 V. Therefore, it is crucial to raise its cutoff voltage to 4.7−4.8 V to achieve high specific energy.[4–6] On the anode side, lithium (Li) metal emerges as an attractive option due to its predominantly high capacity (3860 mAh g^{-1}) and low potential (−3.04 V versus SHE).[7,8] Consequently, high-voltage (≥4.7 V) Li||NCM811 chemistry has been considered as one of the most effective methods to achieve high-energy-density batteries (350−500 Wh kg^{-1}).[9,10]

Unfortunately, most existing electrolytes cannot simultaneously meet the demands for the highly oxidative NCM811 cathode and highly reductive Li metal anode.[11–13] Current Li-ion electrolytes are primarily categorized into two types: carbonates and ethers. Carbonate electrolytes have achieved tremendous success in commercial LIBs with graphite as the anode, and they show promise for high-voltage NCM811 cathodes due to their high stability against oxidization at high potential. However, carbonate electrolytes usually exhibit poor compatibility with Li metal due to the formation of an inferior solid electrolyte interphase (SEI). This causes a continuous electrolyte decomposition and Li dendrite growth, resulting in the dead Li formation, electrolyte dry-up, and ultimately battery failure.[14–16] In contrast, ether-based electrolytes generally exhibit excellent compatibility with the Li metal, owing to their stability against reduction at low potential.[17,18] However, their poor oxidative stability (<4.0 V versus Li/Li^+) significantly limits their applications in high-voltage Li metal batteries (LMBs).[19] It is even more challenging to develop an ether electrolyte that can support

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high-voltage NCM811 cathode at 4.7–4.8 V. Therefore, it is of critical importance to solve the electrolyte "dilemma" to build high-energy Li||NCM811 batteries.

To date, various strategies have been proposed to enhance the high-voltage stability of ether electrolytes, such as high-concentration electrolytes (HCEs, >3 M)[20–22] and localized high-concentration electrolytes (LHCEs, ~2 M).[23–26] HCEs can effectively improve the oxidative stability of ether electrolytes by forming an inorganic-rich cathode–electrolyte interphase,[20,21] while boosting Coulombic efficiency (CE) and cycling life of Li metal anode. However, HCEs introduce additional issues such as substantial cost, high viscosity, and poor electrode wetting.[27] LHCEs are usually designed by adding weakly solvating cosolvents into HCEs, which not only reduce the electrolyte viscosity and cost but also greatly enhance their high-voltage stability.[28] Nevertheless, LHCEs still contain some unstable and uncoordinated free ether solvents, which are prone to oxidization and decomposition at high voltages (≥4.7 V), leading to a rapid performance degradation in high-voltage Li||NCM811 batteries. Recently, researchers have utilized chloro-substituted ethers as solvents to enhance the oxidative stability of LHCEs (up to 4.6 V),[26] but it remains challenging to manage the Li||NCM811 chemistry at 4.7 V or higher. Overall, these ether electrolytes have reached a "4.7 V" bottleneck, which appears to be an insurmountable barrier for high-voltage ether-based Li||NCM811 batteries.[29–31] Therefore, it is indispensable to propose new strategies to design ether molecules and electrolytes that can overcome the 4.7 V limitation and enable stable cycling in high-voltage Li||NCM811 batteries.[32–34]

In this work, we introduce (3,3,3-trifluoropropyl)trimethoxy silane (TTMS) as a multifunctional organic alkoxysilane single-solvent for high-voltage (4.7–4.8 V) and long-cycling Li||NCM811 batteries. This solvent possesses a specialized molecular structure that incorporates both strongly solvating $-OCH_3$ groups and weakly solvating $-CF_3$ segments. The presence of C–F bonds endows intrinsic antioxidation properties for TTMS, leading to a wide electrochemical window of 6 V that far surpasses conventional ether electrolytes. Meanwhile, the strongly solvating $-OCH_3$ groups facilitate the dissolution and high concentration of LiFSI salts, and the designed 2.1 M LiFSI/TTMS electrolyte demonstrates unique solvation structures with abundant contact ion pairs (CIPs) and aggregates (AGGs) within its bulk phase. This desirable solvation structure favors the anion decomposition and generates inorganic-rich interphases on both NCM811 cathode and Li metal anode. Consequently, this electrolyte endows excellent compatibility for the 4.8 V Li||NCM811 battery chemistry, with 84.5% capacity retention achieved after 300 cycles. Even under aggressive conditions, such as high temperature (60 °C) and anode-less configuration (N/P ratio = 1.76), the Li||NCM811 battery still demonstrates remarkable cycling stability, with >80% capacity retention after 300 cycles. Our work indicates that molecular design and engineering is an effective strategy for developing ultrahigh-voltage and long-cycling LMBs.

Results and Discussion

Electrolyte Design and Selection Principle

Conventional ether solvents usually decompose at high potential due to their unstable C–O bonds, whose bonding energy is only ~352 kJ mol^{-1}, lower than C–H bond (~389 kJ mol^{-1}).[35] Moreover, the polarity of C–O bond results in a higher electron density around the O atom, which is susceptible to degradation or decomposition in high-voltage environments. Therefore, the tendency of C–O bond cleavage diminishes the stability of the entire ether molecule, which is the primary cause for its poor oxidation stability.

To tackle this challenge, we adopted a two-step molecule design approach to introduce both Si–O and C–F bonds to the ether molecule. The Si–O bond offers much higher energy (~480 kJ mol^{-1}) than the C–O bond (352 kJ mol^{-1});[36] therefore, if we replace one C atom in the ether (C–O–C) with a Si atom, we can obtain a silyl ether molecule (C–O–Si) with improved oxidation stability. Moreover, the C–F bond is one of the strongest single bonds in organic compounds, which has significantly higher energy (527 kJ mol^{-1}) than the C–O bond.[37] Due to the high electronegativity of the F atom, it exerts a strong inductive effect that causes a higher electron density around itself. Consequently, the fluorination of ether molecules leads to a shift in the molecular electron cloud, further enhancing the molecule's oxidation stability. In this work, we aim to explore the synergistic effect between Si–O and C–F bonds to boost the molecular stability and identify the optimal electrolyte for high-voltage Li||NCM811 batteries.

To obtain antioxidization ether molecules, our molecule design approach incorporates Si–O bonds first, followed by the introduction of C–F bonds (Figure 1a). The starting molecule is methoxytrimethylsilane (MTMS) with one Si–O bond. By increasing the number of Si–O bonds, we obtain dimethyldimethoxysilane (DDMS) with two Si–O bonds and methyltrimethoxysilane (MTTS) with three Si–O bonds. Lastly, we extended the carbon chains of these silyl ether molecules and fully fluorinated them, yielding (3,3,3-trifluoropropyl)dimethylmethoxysilane (TDMS), (3,3,3-trifluoropropyl)methyldimethoxysilane (TMDS), and TTMS.

We utilized theoretical methods to simulate molecular stability against reduction and oxidization. According to frontier orbital theory, the energy levels of a molecule's highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) can be used to predict its oxidative and reductive stability, respectively.[38–40] Thus, density functional theory (DFT) calculations were performed on six silyl ethers, two common carbonate solvents: ethylene carbonate (EC) and dimethyl carbonate (DMC), and two conventional ethers (Figure 1b and Table S1).

MTMS and DDMS, which contain one and two Si–O bonds, respectively, have the highest HOMO energy level (−7.31 eV), indicating poor oxidative stability. However, when three Si–O bonds are introduced, MTTS exhibits a significantly lower HOMO energy level (−7.46 eV), suggesting its improved oxidative stability compared to MTMS

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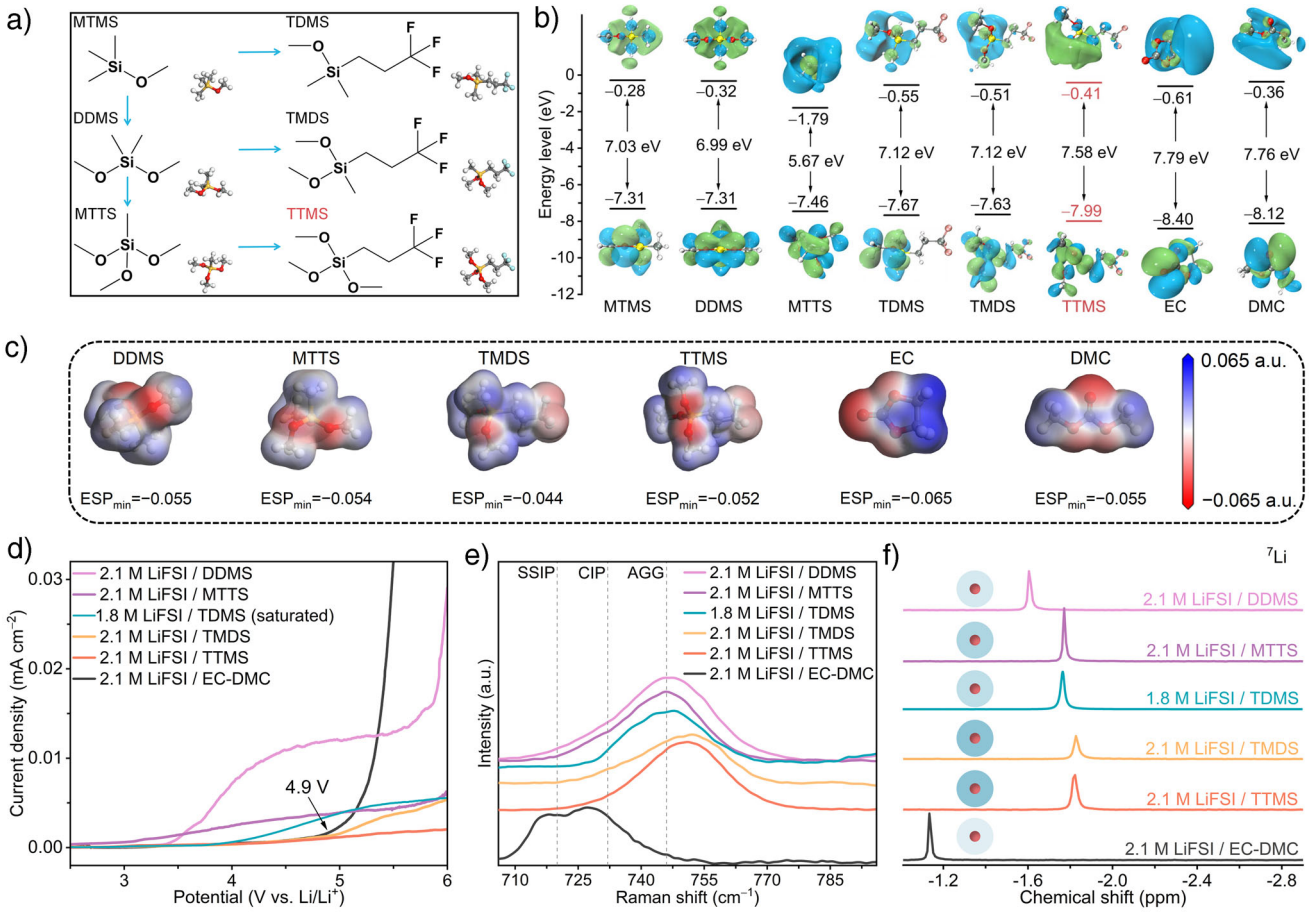


Figure 1. Design and characterization of silyl ether solvents. a) The scheme of the two-step molecule design approach. b) Calculated HOMO and LUMO energy levels of different solvent molecules. c) Calculated electrostatic potential (ESP) maps of solvents. d) LSV measurements of Li||Al cells with different electrolytes at scan rates of 0.1 mV s^{-1}. e) Raman spectra, f) ^{7}Li NMR spectra of different electrolytes.

and DDMS. Despite this, MTTS shows the lowest LUMO energy level (−1.79 eV), indicating its poor reductive stability. Therefore, none of these unfluorinated silyl ethers are suitable as electrolyte solvents.

In contrast, fluorinated TMDS and TDMS exhibit significantly lower HOMO energy levels (−7.67 and −7.63 eV, respectively), along with comparable LUMO energy levels (−0.55 and −0.51 eV), indicating improved oxidative and reductive stability. Remarkably, TTMS, the fluorinated version of MTTS, shows a substantial decrease in its HOMO energy (−7.99 eV versus −7.46 eV) while exhibiting a notable increase in its LUMO energy (−0.41 eV versus −1.79 eV). This suggests that fluorination dramatically enhances both oxidative and reductive stability of TTMS. Even compared to the common carbonate solvent (DMC), TTMS has similar HOMO and LUMO energy levels (−7.99 eV versus −8.12 eV and −0.41 eV versus −0.36 eV, respectively), underscoring its potential as an antioxidization solvent for high-voltage Li metal batteries. Moreover, TTMS exhibits a much higher ESP_{min} (ESP_{min}: electrostatic potential minimum) value than EC (Figure 1c), indicating its weaker solvation ability. This characteristic facilitates the desolvation process of Li^+ ions in the electrolyte, thereby favoring the formation of the EEIs enriched with inorganic components.

To further evaluate the properties of these silyl ethers, they were formulated with lithium bis(fluorosulfonyl)imide (LiFSI) into single-solvent electrolytes. The LiFSI concentration was tentatively set at 2.1 M, a moderate concentration that is close to LHCEs. However, the TDMS solvent only contains one C─O─Si group, leading to its moderate solvation capability toward lithium salts. The maximal concentration is 1.8 M LiFSI in TDMS, which is compared with other 2.1 M concentration electrolytes. Linear sweep voltammetry (LSV) was used to examine the oxidation stability of these electrolytes (Figure 1d and Table S2). Among them, the 2.1 M LiFSI/TTMS exhibits the best stability (>6 V), far exceeding the 2.1 M LiFSI/EC-DMC carbonate electrolyte (4.9 V). This superior performance also indicates the inhibited Al collector corrosion in LiFSI/TTMS. Besides, LiFSI/TDMS and LiFSI/TMDS electrolytes also demonstrate reasonable oxidation stability, suggesting the effectiveness of the fluorinated silyl ethers molecular design.

The electrochemical properties of electrolytes are closely related to their solvation structures. Conventional Li-ion electrolytes typically comprise strong solvating solvents and low-concentration lithium salts. In this case, their solvation structure is dominated by separated solvent ion pairs (SSIPs), where Li^+ ions are mainly coordinated with the solvent,

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whereas anions are excluded from the first solvation shell of Li^+.[41] However, when a weaker solvating solvent is used, Li^+ ions tend to coordinate more with anions, forming a large number of CIPs or even AGGs. The increment of CIPs and AGGs facilitates the preferential anion decomposition at the electrode interface, generating a high-quality electrode–electrolyte interphase that is rich in inorganic materials.[42,43] This dense interphase further helps isolate the reactive electrode from the bulk electrolyte and reduces electrolyte decomposition side reactions, thus enhancing the battery operation voltage and cycling stability.[44]

We employed Raman spectra and nuclear magnetic resonance (NMR) to further understand the solvation structures of these electrolytes. The S—N—S bond in FSI^- anions usually shows a vibration peak in the 710–760 cm^{-1} range, and its peak position shift suggests different interactions with Li^+ (Figures 1e and S1).[45,46] In the 2.1 M LiFSI/EC-DMC electrolyte, the Raman peak appears at ~722 cm^{-1}, suggesting that a large number of SSIPs are present. In comparison, the silyl ether electrolytes exhibit varying degrees of blue shifts. For DDMS-, MTTS-, and TDMS-based electrolytes, their S—N—S peaks shift to ~743 and 742 cm^{-1}, respectively, suggesting stronger interactions between Li^+ cations and FSI^- anions. This observation implies that these two electrolytes are dominated by CIPs, with AGGs as a minor fraction. The largest blue shift occurs in the LiFSI/TMDS and LiFSI/TTMS electrolytes (~749 cm^{-1}), suggesting that their solvation structures are mainly composed of CIPs and AGGs.

The ^7Li NMR spectra further corroborated the Raman results (Figure 1f). Compared to LiFSI/EC-DMC, LiFSI/DDMS, LiFSI/MTTS, and LiFSI/TDMS electrolytes exhibit larger up-field shifts (lower chemical shifts), while LiFSI/TMDS and LiFSI/TTMS show even more pronounced up-field shifts, suggesting a progressively stronger shielding effect on Li^+ ions.[47–49] These results reveal that, from LiFSI/EC-DMC to LiFSI/TTMS, the interactions between Li^+ and FSI^- gradually intensify, with more and more FSI^- ions entering the first solvation shell of Li^+.

In summary, the LiFSI/TTMS electrolyte system not only demonstrates optimal oxidative stability but also features AGG-rich solvation structures. Additionally, compared to other fluorinated alkoxysilanes and prior-reported solvents for 4.7 V cells, TTMS features a lower cost and higher thermal stability (Table S3). Therefore, TTMS is identified as the optimal electrolyte solvent.

To identify the optimal LiFSI concentration, we explored the effect of LiFSI concentrations (1–2.4 M) on the electrolyte conductivity and solvation structures. As shown in Figure S2, the electrolyte conductivity initially increases and then decreases with increasing salt concentration, with the highest conductivity (0.586 mS cm^{-1}) achieved at 2.1 M. Raman analysis (Figure S3) suggests the presence of CIPs and AGGs in these LiFSI-TTMS electrolytes. NMR spectra (Figures S4 and S5) further indicate that 2.1 M is the critical concentration to achieve a near-saturated solvation structure that is comparable to the 2.4 M electrolyte. Therefore, 2.1 M LiFSI is identified as an optimal concentration due to its moderate concentration, highest ionic conductivity, and desirable solvation structures with abundant CIPs and AGGs.

MD Simulations for Electrolytes

Molecular dynamics (MD) simulations are conducted to gain deeper insight into the solvation structure of 2.1 M LiFSI/TTMS. For better comparison, we also evaluated the standard 1 M $LiPF_6$/EC-DMC and 2.1 M LiSFI/EC-DMC electrolytes. Global snapshots of these electrolytes are shown in Figure 2a–c. In the 1 M $LiPF_6$/EC-DMC electrolyte, the distance between Li^+ and PF_6^- is notably large, suggesting weak cation–anion coordination. Instead, Li^+ preferentially coordinates with carbonate solvents, as seen in a typical local snapshot (Figure 2d). The primary Li-ion solvation shell consists of two EC and two DMC molecules, with no PF_6^- present. This coordination structure is further confirmed by the radial distribution function (RDF, g(r)) and coordination number (CN) plots for Li^+ (Figure 2g). The g(r) for Li^+–O in EC-DMC is larger, with an average CN of 4.2 (2.9 Å). In contrast, the g(r) for Li^+–F in PF_6^- is smaller, and the average CN is only 0.4. This suggests that PF_6^- cannot easily enter the first solvation shell of Li^+, resulting in a solvation environment that is predominantly composed of SSIPs.

In the 2.1 M LiFSI/EC-DMC electrolyte, the solvation structure has moderately changed compared with the standard 1 M electrolyte. Although the distribution of Li^+ and FSI^- remains relatively loose in the global snapshot (Figure 2b), the typical local snapshot shows that one FSI^- ion has entered the first solvation shell of Li^+ (Figure 2e), likely due to the higher concentrations and enhanced cation–anion interactions. Moreover, the average CN for Li^+–O in EC-DMC slightly decreases from 4.2 to 3.6, while the average CN for Li^+–O in FSI^- is 1.0, indicating that a small portion of FSI^- has entered the Li^+ solvation shell and formed some CIPs. Despite this, the solvation environment of 2.1 M LiFSI/EC-DMC is still dominated by SSIPs, which agrees with the prior Raman results (Figure 1e).

In contrast, the 2.1 M LiFSI/TTMS electrolyte exhibits a vastly different solvation structure compared with 2.1 M LiFSI/EC-DMC, despite their identical salt concentrations. In the global snapshot (Figure 2c), Li^+ and FSI^- ions are tightly bound and closely aggregated. Moreover, the typical local snapshot (Figure 2f) shows that the first solvation shell of Li^+ contains three FSI^- anions and one TTMS molecule, which is characteristic of AGGs. Furthermore, the average CN for Li^+–O in TTMS is small (only 0.9), while the average CN for Li^+–O in FSI^- is considerably larger (3.6). This indicates that most FSI^- ions have entered the first solvation shell of Li^+, forming a substantial number of AGGs structures. These results are in good agreement with the Raman and NMR results.

We believe that the solvation differences between 2.1 M LiFSI/EC-DMC and 2.1 M LiFSI/TTMS are attributed to their distinct solvent properties. The EC/DMC solvents contain strongly solvating carbonate groups (—C=O), which exhibit a high affinity for binding with Li^+ cations. At the moderately high concentration (2.1 M), some Li^+ ions interact with FSI^- anions; however, most Li^+ ions remain coordinated by EC/DMC molecules, resulting in SSIPs as the major solvation structure. In contrast, TTMS possesses an electron-withdrawing —CF_3 segment, which not only results

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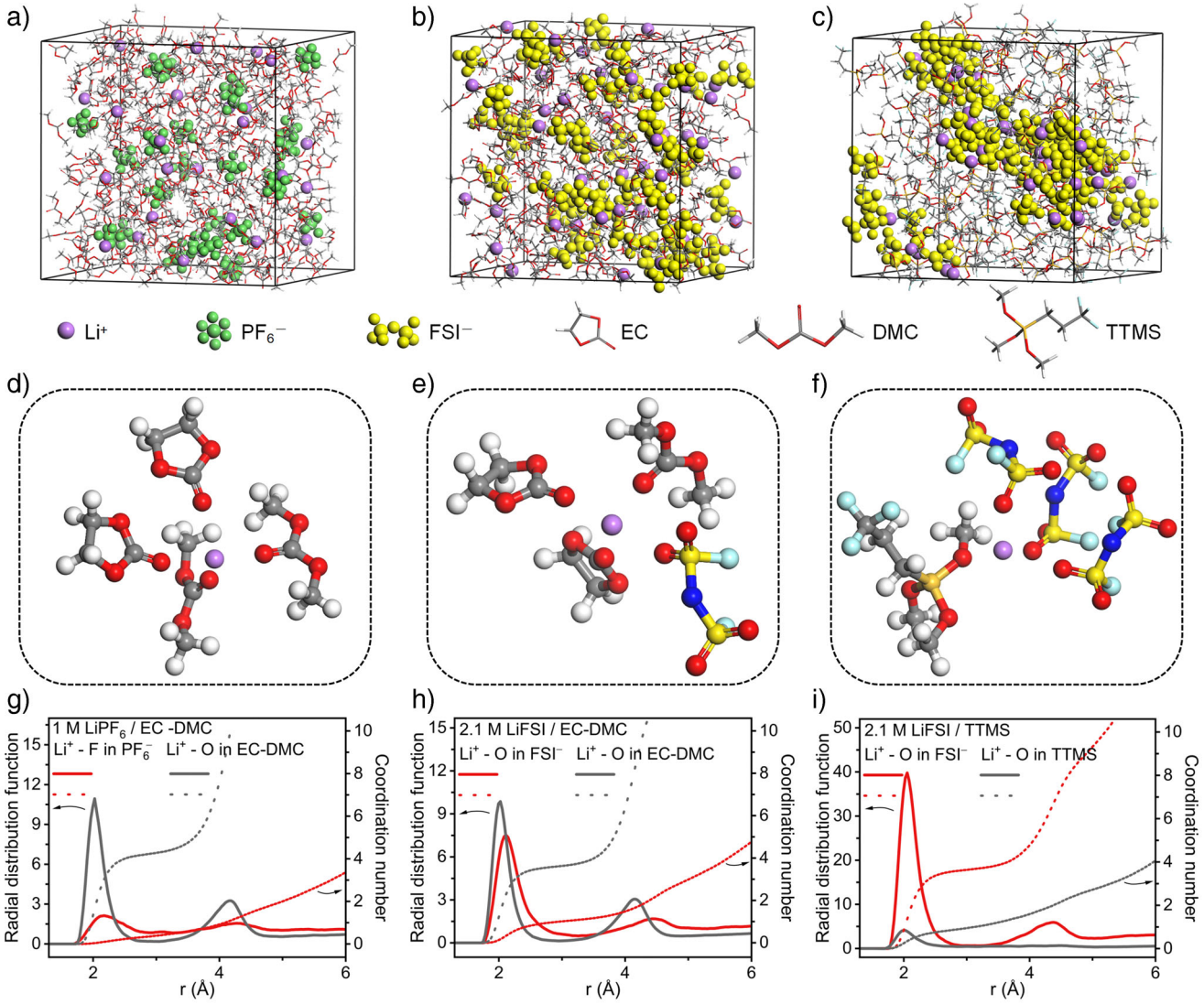


Figure 2. MD simulations of the solvation behavior in different electrolytes. a)–c) Global snapshots with Li^+ and anions highlighted in (a) 1 M $LiPF_6$/EC-DMC, (b) 2.1 M LiFSI/EC-DMC, and (c) 2.1 M LiFSI/TTMS. d)–f) Representative local snapshots in (d) 1 M $LiPF_6$/EC-DMC, (e) 2.1 M LiFSI/EC-DMC, and (f) 2.1 M LiFSI/TTMS. g)–i) Calculated RDF (g(r)) and CN in (g) 1 M $LiPF_6$/EC-DMC, (h) 2.1 M LiFSI/EC-DMC, and (i) 2.1 M LiFSI/TTMS.

in a more uniform electron distribution and a higher ESP_{min} but also introduces notable steric hindrance effects. These combined factors collectively mitigate the binding propensity between Li^+ and TTMS, thereby limiting the coordination to a maximum of two TTMS within the first solvation shell of Li^+ (Figure S6). As a result, the interactions between Li^+ and FSI^- get strengthened, promoting the formation of CIPs and AGGs as the dominant solvation structures.

Compatibility and Plating/Stripping Reversibility with Lithium

Electrolytes that contain abundant CIPs and AGGs can facilitate the formation of a dense and high-quality SEI, thereby inhibiting Li dendrite growth and improving the Li plating efficiency.[50,51] To validate this conjecture, we first conducted plating/stripping tests using Aurbach's method. To understand the effect of LiFSI concentrations on Li metal performance, we assembled Li||Cu cells with various concentrations (1.0–2.4 M). As shown in Figures 3a and S7, 2.1 M LiFSI/TTMS achieves the highest average Coulombic efficiency (CE = 99.21%). At lower concentrations (1 and 1.5 M), the average CE is below 99%. Conversely, at 2.4 M, the average CE slightly decreases to 99.19%, possibly due to the higher concentration and increased electrolyte viscosity. Therefore, 2.1 M is identified as the optimal concentration for Li metal anode.

The 2.1 M LiFSI/TTMS electrolyte also provides superior performance to 1 M $LiPF_6$/EC-DMC and 2.1 M LiFSI/EC-DMC electrolytes. These two carbonate electrolytes suffer from significant potential fluctuations (even exceeding 4 V) and cannot even complete the Aurbach's test (Figure 3b). We also used the conventional method to evaluate the plating efficiency and cycling life of Li||Cu batteries (Figure 3c). Over

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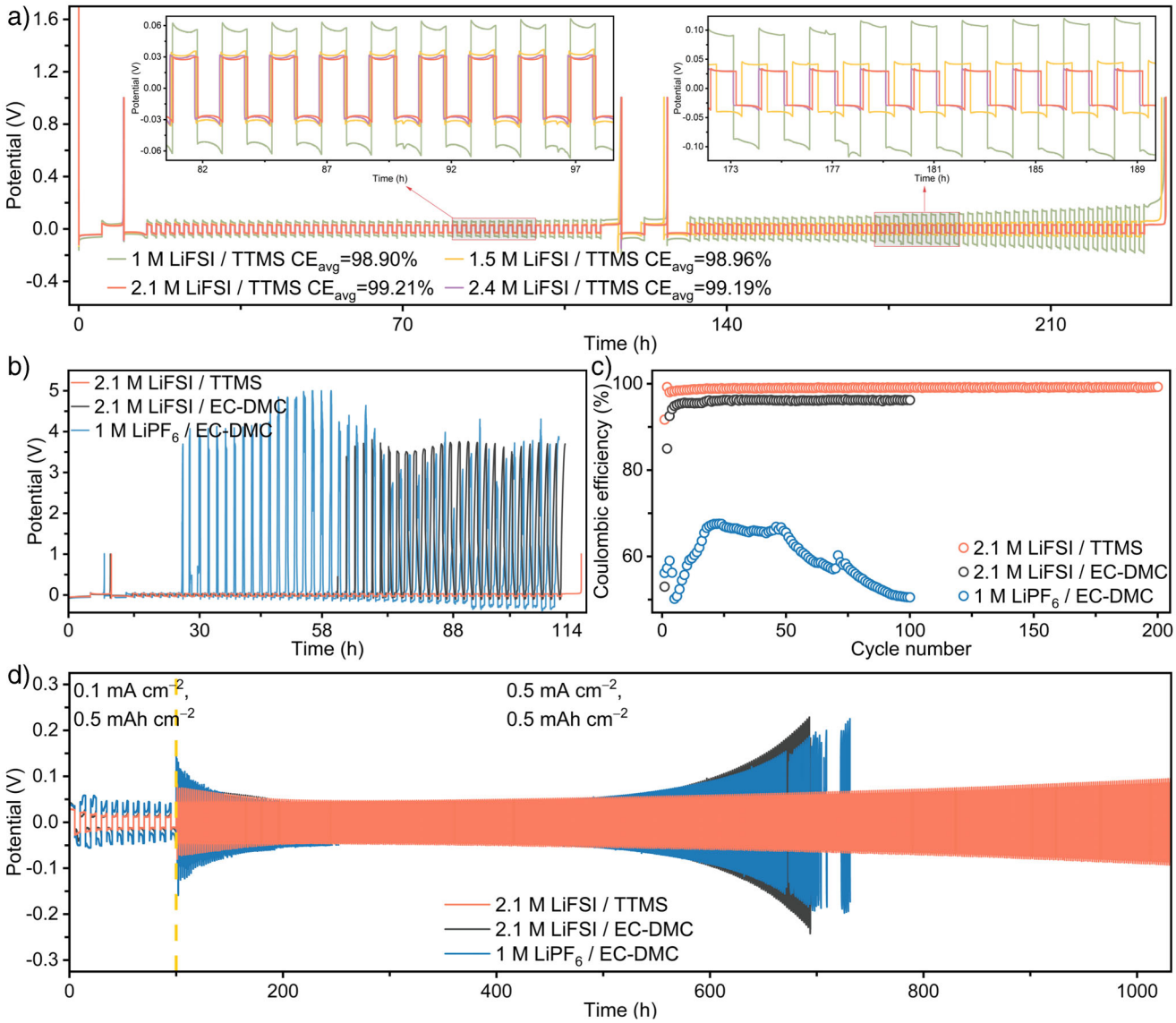


Figure 3. Compatibility and reversibility of plating/stripping of different electrolytes with Li metal. a) and b) Aurbach's tests at current densities of 0.5 mA cm^{-2} in Li||Cu cells with varying electrolytes. c) Conventional CE tested at current densities of 0.2 mA cm^{-2} in Li||Cu cells with different electrolytes. d) Polarization curves of Li||Li cells with different electrolytes.

100 cycles, the average CE for 1 M $LiPF_6$/EC-DMC and 2.1 M LiFSI/EC-DMC is ~60% and ~96%, respectively, which is much lower than that of 2.1 M LiFSI/TTMS (99.1% over 200 cycles, Figures S8–S10). This efficiency is also comparable to that in the well-studied LHCE of LiFSI/1.2 DME/3TTE (Figures S11 and S12).

We also assembled Li||Li symmetric cells to further explore the electrolyte compatibility with Li metal. These cells were activated at 0.1 mA cm^{-2} for 100 h and then tested at a higher current density (0.5 mA cm^{-2}) to compare the potential changes over time (Figure 3d). Remarkably, the 2.1 M LiFSI/TTMS electrolyte not only leads to the smallest overpotential but also shows the most stable cycling performance (1000 h). By contrast, the Li||Li batteries with carbonate electrolytes suffer from drastic voltage fluctuations and cell short circuits after ~700 h.

To understand the Li anode performance differences in these electrolytes, we disassembled Li||Cu cells after ~50 cycles, retrieved Cu and Li foils, and conducted scanning electron microscopy (SEM) and XPS spectra analyses on both Cu and Li foils. As shown in Figure 4a,b, the Cu foil surfaces exhibit uneven, porous, and sparse fibrous characteristics in 1 M $LiPF_6$/EC-DMC ($LiPF_6$/EC-DMC) and 2.1 M LiFSI/EC-DMC (LiFSI/EC-DMC). In contrast, the Cu foil in 2.1 M LiFSI/TTMS (LiFSI/TTMS) shows a much more uniform and denser deposition morphology (Figure 4c). On the other hand, Li foils cycled in both carbonate electrolytes are severely corroded by electrolytes, with large cracks and some Li dendrites observed (Figure 4d,e). Remarkably, the Li foil in 2.1 M LiFSI/TTMS is uniform, dense, and intact, without any dendrite formation (Figure 4f).

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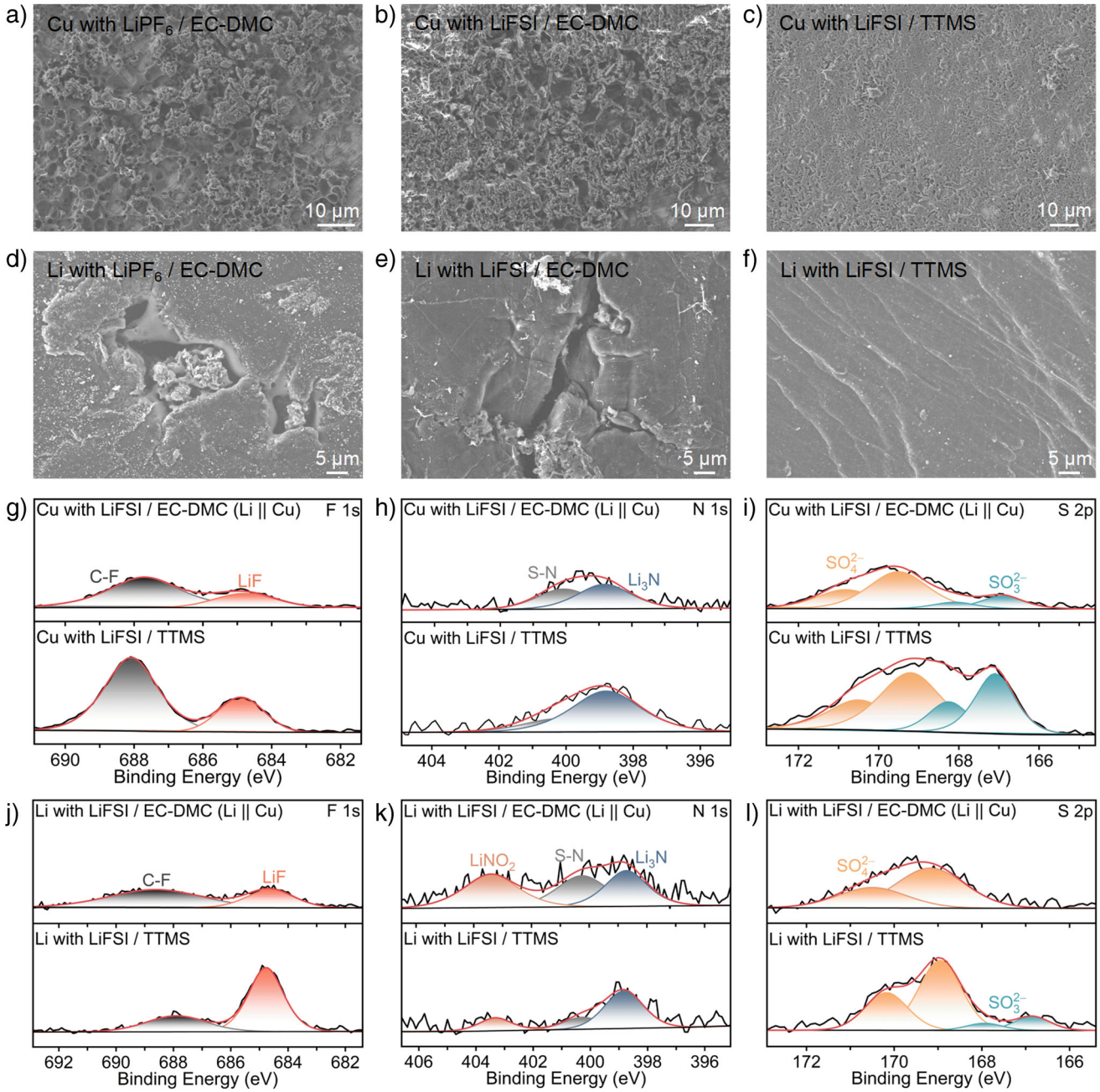


Figure 4. Microscopic morphology and SEI properties of Li|| Cu cells with different electrolytes. a)–f) SEM images of Cu and Li foils from cycled Li||Cu cells with (a and d) 1 M $LiPF_6$/EC-DMC, (b and e) 2.1 M LiFSI/EC-DMC, and (c and f) 2.1 M LiFSI/TTMS. g) and j) F 1s, h) and k) N 1s, and i) and l) S 2p spectra derived from the XPS spectra of Cu and Li foils of Li||Cu cells cycled with different electrolytes.

Furthermore, X-ray photoelectron spectroscopy (XPS) reveals more details for the SEI composition of Li and Cu electrodes with different electrolytes (Figures 4g–l and S13–S15). Both Li and Cu electrodes with 2.1 M LiFSI/TTMS exhibit strong signals for substances like LiF, Li_3N, Li_2SO_4, and Li_2SO_3, indicating an inorganic-rich SEI composition. As discussed in Raman and NMR results, the TTMS electrolyte exhibits a desirable solvation structure with CIPs and AGGs, which can promote the FSI^- anion decomposition, generating inorganic-rich SEI components. This SEI layer plays a crucial role in suppressing the Li dendrite growth and achieving uniform Li plating. These favorable morphological and SEI properties well explain the high efficiency (~99.2%) and long cycling (1000 h) of Li metal anode with the TTMS electrolyte.

Performance and Characterization of Li||NCM811 Cells with Different Electrolytes

We assembled Li||NCM811 batteries to evaluate the TTMS electrolyte stability against high voltages (4.7 V, Figures 5a and S16–S18). Interestingly, the 2.1 M LiFSI/EC-DMC elec-

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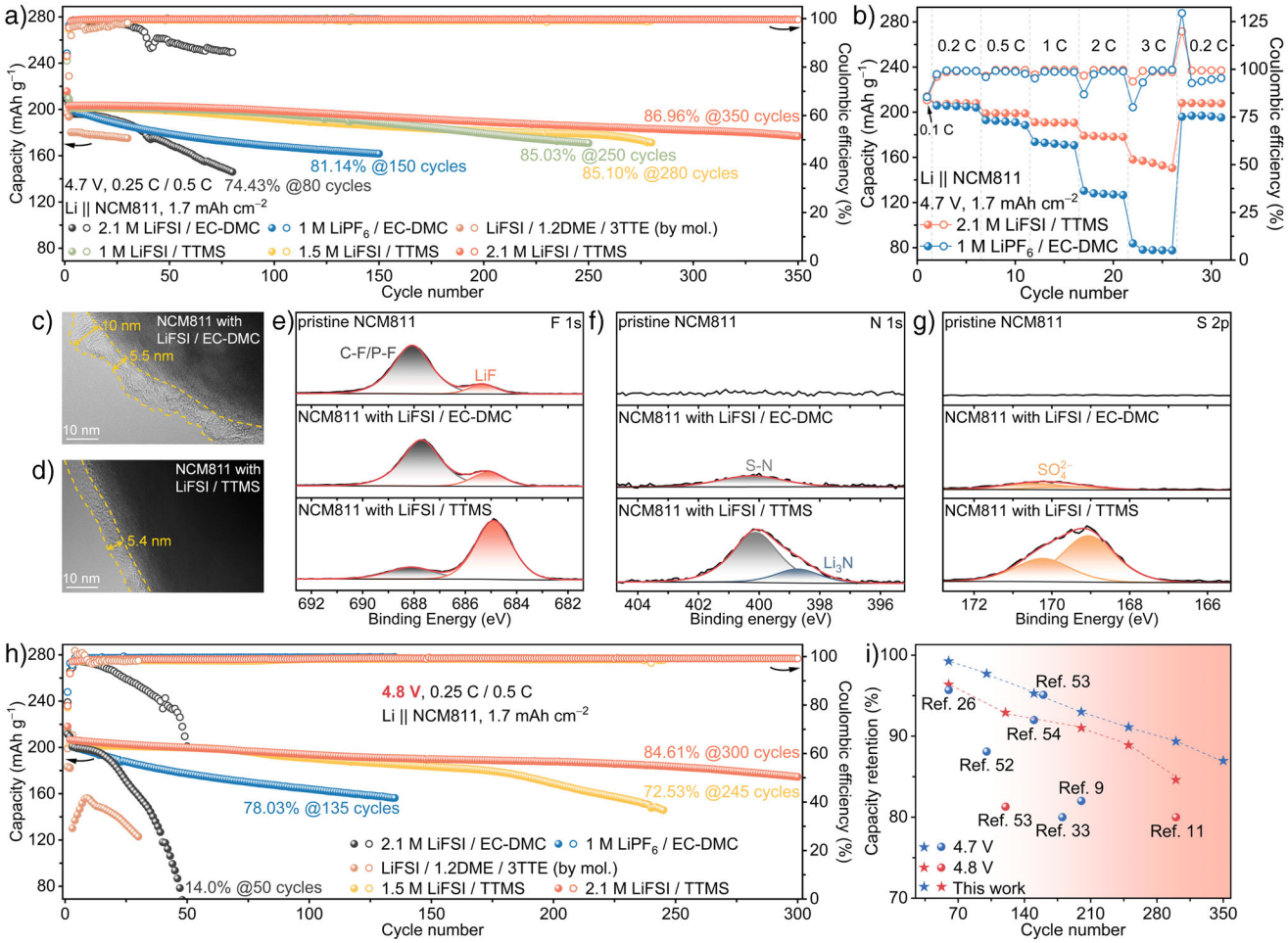


Figure 5. Performance and characterization of Li||NCM811 cells with different electrolytes. a) Long-term performance of Li||NCM811 cells with different electrolytes at 4.7 V. b) Rate performance of Li||NCM811 cells with 1 M $LiPF_6$/EC-DMC and 2.1 M LiFSI/TTMS. c) and d) TEM images of cycled NCM811 at 4.7 V with (c) 2.1 M LiFSI/EC-DMC and (d) 2.1 M LiFSI/TTMS. e) F 1s, f) N 1s, and g) S 2p spectra derived from the XPS spectra of the CEI on NCM811 cycled with different electrolytes at 4.7 V. h) Long-term performance of Li||NCM811 cells with different electrolytes at 4.8 V. i) Comparison of the cycling stability of Li||NCM811 cells with 2.1 M LiFSI/TTMS in this work with previous reports.

trolyte exhibits the worst cycling performance (~140 mAh g^{-1}@80 cycles), which is even worse than the standard 1 M $LiPF_6$/EC-DMC electrolyte, likely due to the aluminum foil corrosion by FSI^- anions at high voltages. Electrochemical impedance spectroscopy (EIS) (Figure S19) and SEM analysis (Figure S20) confirmed the electrode corrosion issue. For the LHCE electrolyte of LiFSI/1.2DME/3TTE, the NCM811 battery delivers a low discharge capacity, possibly due to the electrolyte oxidization at high voltages. In contrast, even the 1 M LiFSI/TTMS electrolyte can overcome the electrode corrosion or electrolyte decomposition issue, and the NCM811 battery can operate stably for more than 200 cycles with a promising capacity retention of 88.64%. The 2.1 M LiFSI/TTMS electrolyte leads to the best cycling stability for NCM811 battery, which retains 86.96% of its capacity at 350th cycle and a high average CE of 99.68%. To our knowledge, this is also one of the best cycling performances reported on Li||NCM811 batteries at 4.7 V (Figure 5i, Tables S4 and S6).[52–54]

Besides the cycling stability, the 2.1 M LiFSI/TTMS electrolyte also results in high-rate performance for NCM811 batteries, where the discharge capacities are 214, 208, 199, 190, 178, and 157 mAh g^{-1} at 0.1 C, 0.2 C, 0.5 C, 1 C, 2 C, and 3 C, respectively (Figures 5b and S21). Remarkably, this rate performance is even better than the standard 1 M $LiPF_6$/EC-DMC electrolyte, despite its well-known high conductivity (~10 mS cm^{-1}). The inferior rate performance in 1 M $LiPF_6$/EC-DMC should result from the poor cathode–electrolyte interphase (CEI) on the cathode and low-quality SEI on the Li metal anode, leading to high interfacial resistance and slow Li^+ ion conduction (Figure S19).

To compare the CEI difference in different electrolytes, we conducted transmission electron microscope (TEM) and XPS studies. As shown in Figure 5c and Figure S22, the CEI layer derived from 2.1 M LiFSI/TTMS is uniform, compact, and thin (~5.4 nm), which is superior to that in 2.1 M LiFSI/EC-DMC. Such a thin and dense CEI layer plays a crucial role in enhancing the interfacial stability and cycling stability of NCM811 cathodes at high voltages. Besides, these two electrolytes exhibit significant differences in the chemical compositions of CEI layers. In 2.1 M LiFSI/TTMS,

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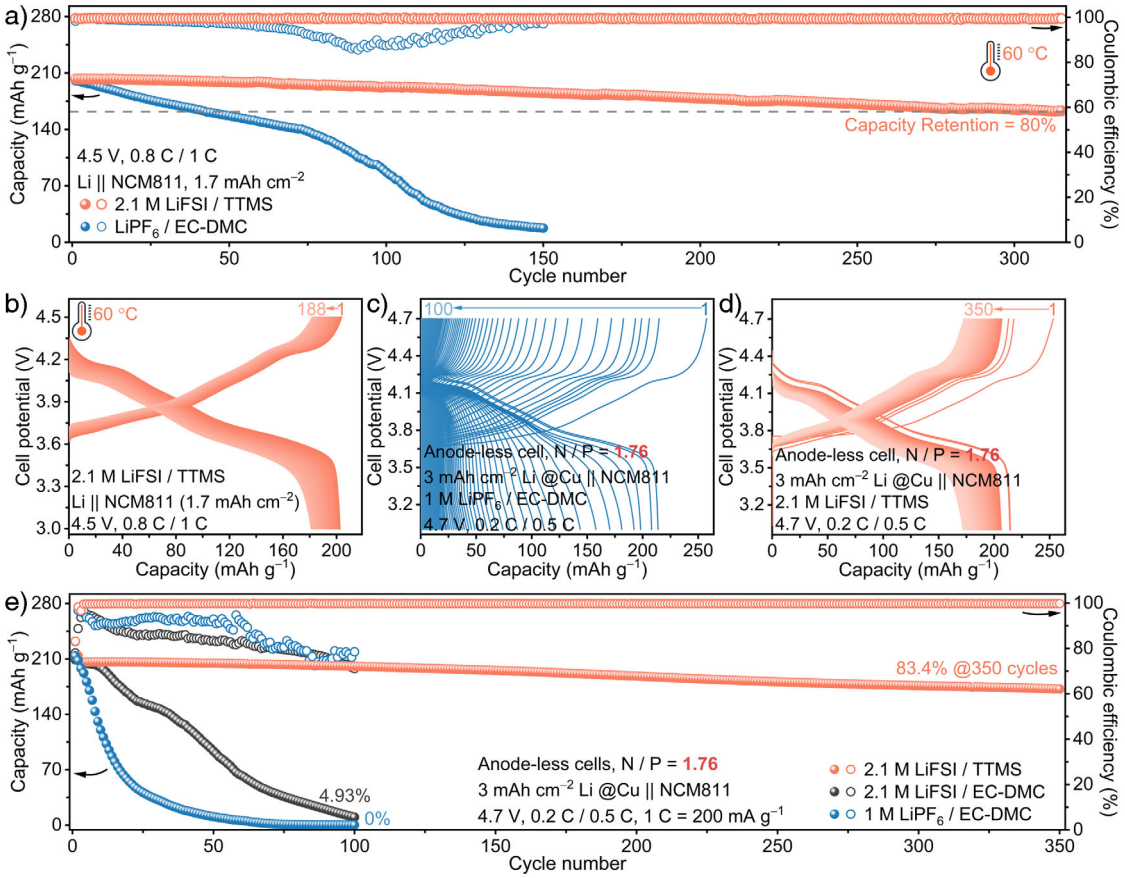


Figure 6. a) Long-term performance of Li||NCM811 cells with 1 M $LiPF_6$/EC-DMC and 2.1 M LiFSI/TTMS at 60 °C. b) Potential-capacity profiles of the Li||NCM811 cell with 2.1 M LiFSI/TTMS at 60 °C. c) and d) Potential-capacity profiles of anode-less Li||NCM811 cells with (c) 1 M $LiPF_6$/EC-DMC and (d) 2.1 M LiFSI/TTMS at 4.7 V. e) Long-term performance of anode-less Li||NCM811 cells with different electrolytes at 4.7 V and 25 °C.

the CEI consists of more elements like Li, F, N, S, and O but contains less C elements, indicating its inorganic-rich CEI properties (Figures S23–S25). This observation is further supported by the F 1s, N 1s, and S 2p spectra (Figure 5e–g). In the F 1s spectra, the CEI from LiFSI/TTMS exhibits a higher intensity of LiF, whereas 2.1 M LiFSI/EC-DMC shows a lower intensity of LiF but a higher intensity of C─F. Moreover, we also observe higher intensities of S─N and Li_3N in the N 1s and SO_4^{2-} in the S 2p spectra for the 2.1 M LiFSI/TTMS.[37] These inorganic components result from the anion decomposition reactions, which can effectively enhance the interfacial stability and cycling performance of NCM811.

We further evaluated the Li||NCM811 battery performance at an ultrahigh cut-off voltage of 4.8 V, which is rarely studied in most ether electrolyte studies. As shown in Figures 5h and S26, the 2.1 M LiFSI/EC-DMC electrolyte still exhibits the worst cycling performance due to the electrode corrosion issue. The 1 M $LiPF_6$/EC-DMC electrolyte only shows a capacity retention of ~78.03% after 135 cycles. By contrast, the 2.1 M LiFSI/TTMS electrolyte demonstrates the best capacity retention (84.61%) and highest average CE (99.08%) after 300 cycles, which is also the best performance among all previous reports (Figure 5i, Tables S5 and S6).

Performance of Li||NCM811 Cells at 60 °C and Anode-Less Full-Cells

It is of practical importance to evaluate the Li||NCM811 battery stability at high-temperature conditions such as 60 °C. We fabricated Li||NCM811 cells with 1 M $LiPF_6$/EC-DMC and 2.1 M LiFSI/TTMS electrolytes, whose cut-off voltage is set as 4.5 V (Figures 6a,b and S27). Remarkably, the cell with 2.1 M LiFSI/TTMS maintains stable cycling over 300 cycles, with a capacity retention of ~80% and a high average CE of 99.47%. As far as we know, this result represents the best cycling performance for Li||NCM811 cells at 60 °C compared to previous studies.

Reducing the Li metal mass is an effective method to improve the energy density and safety of practical Li metal batteries. Therefore, we used 3 mAh cm^{-2} Li@Cu as the anode (Figure S28) and assembled Li||NCM811 full cells with a controlled N/P ratio = 1.76. As shown in Figure 6c–e, the full cells with 1 M $LiPF_6$/EC-DMC and 2.1 M LiFSI/EC-DMC rapidly decay to almost zero within 100 cycles (Figure S29), but the 2.1 M LiFSI/TTMS electrolyte exhibits exceptional cycling life and achieves a capacity retention of ~83.4% after 350 cycles. The excellent cycling performance in the Li||NCM811

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full cell is attributed to the interfacial compatibility of TTMS electrolyte to both high-voltage NCM811 cathode and low-potential Li metal anode, as discussed in the half-cell studies (Figures 4 and 5).

To further explore the feasibility of the TTMS electrolyte, we also fabricated a graphite||NCM811 pouch battery with a capacity of 1 Ah. This pouch battery only underwent a simple electrolyte injection and sealing process, without any degassing or second sealing treatment. Impressively, it achieves a long cycling life of 600 cycles, with ~80% capacity retention and an average CE of 99.81% (Figure S30). In addition, there are no signs of gas generation, bulging, or swelling in the pouch battery after 600 cycles (Figures S31 and S32). This indicates that the 2.1 M LiFSI/TTMS electrolyte is not only promising for high-voltage Li metal batteries but also compatible with graphite-based Li-ion batteries, which holds great potential for practical applications.

Conclusion

In summary, we propose TTMS as a multifunctional organic silane single solvent for high-voltage Li metal batteries. This solvent exhibits a unique molecular structure, with both strong solvating $-OCH_3$ groups and a weakly solvating $-CF_3$ segment. Therefore, the designed 2.1 M LiFSI/TTMS electrolyte forms favorable solvation structures with abundant CIPs and AGGs within its bulk electrolyte. This solvation structure promotes the anion decomposition and yields inorganic-rich interphases on both high-voltage NCM811 cathode and low-potential Li metal anode. Consequently, this electrolyte exhibits remarkably high oxidative stability (>6 V) and excellent electrode compatibility, leading to stable Li||NCM811 batteries under harsh conditions, such as ultra-high voltage (4.7–4.8 V), high temperature 60 °C, and limited N/P ratio (1.76). Our work provides new insights into the electrolyte design and engineering, paving the way for ultra-high-voltage Li metal batteries.

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Conflict of Interests

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the Supporting Information of this article.

Keywords: (3,3,3-Trifluoropropyl)trimethoxy silane · Electrode–electrolyte interface · Li-metal batteries · Solvation structure · Ultra-high voltage

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